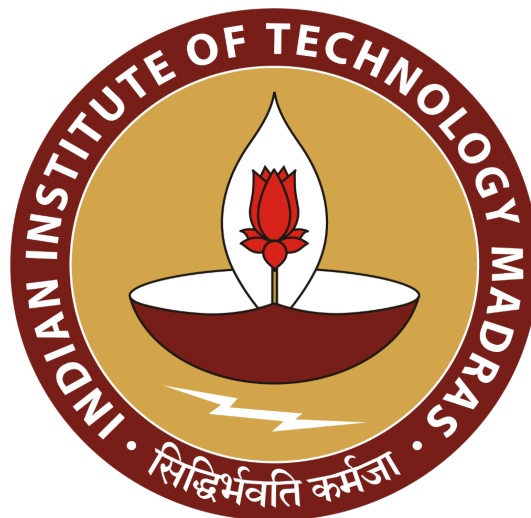


CASE STUDY OF A KIRANA STORE : PROBLEMS, STRATEGIES AND APPROACH

**Final report for the BDM Capstone Project
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1.Executive Summary:

This project report serves as the final report for the Capstone project that is being done as a part of the Business Data Management course. This report focuses on the data analysis of purchase and sales data of a retail business , Kanish Stores, a Grocery shop and explains in detail about the analysis techniques used for finding insights and recommendations for the business which could help them to grow their business.

Kanish Stores is a B2C business specializing in the distribution of various goods, including milk, vegetables, masala items, and some non-grocery products. The store procures these items from wholesale suppliers on a daily basis and sells them directly to customers.

Even though they are in the business for over 6 years now, the business faces hardships in understanding the customer's demand. They are not able to increase their profitability as well as their sales volume. They had to bear losses due to some of the non-grocery products they sold.

The issues will be addressed by analyzing the historical data of both the purchase and sales records from the business, demand forecasting, time series analysis to analyse

the profit made in the last financial year and analyzing the ratio of grocery to non-grocery products

The data analysis for this project was conducted using both Google Sheets and Python. The link to the spreadsheet and the Colab notebook utilized for the analysis is provided in the report for reference. The expected outcome of this project is to provide the business with enhanced insights into sales and purchase reports derived from past financial transactions. This will enable a deeper understanding of customer demand patterns, thereby facilitating better decision-making.

2.Data Analysis Process

a.Data Collection

The data required for analysis is collected for a period of 3 months from the business. All transaction information was manually recorded in a notebook . The handwritten entries were transferred to spreadsheets for better analysis .Tableau is used to analyze the data.

Link to Data:[original_sales_data.xlsx](#)

Link to Colab:  [BDM.ipynb](#)

b. Data Description And Metadata

Two categories of data such as sales and purchase were collected from the business. After preprocessing of data, 11 columns with 592 entries in sales and 9 columns with 592 entries in the purchase category respectively.

1. Sales:

Sales transactions represent the store's revenue, reflecting the goods sold to customers or other retail shops.

2. Purchase:

Purchase transactions track the procurement of goods from wholesale stores which are essential for the store's daily operations. These contribute to the store's inventory and impact its sales and profitability.

```

RangeIndex: 593 entries, 0 to 592
Data columns (total 11 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Date                 593 non-null    datetime64[ns]
1   Item Name            593 non-null    object
2   Category             593 non-null    object
3   Price                593 non-null    int64
4   Quantity             593 non-null    int64
5   Total Cost           593 non-null    int64
6   Profit/Loss          593 non-null    int64
7   Transaction Type     593 non-null    object
8   Week No              593 non-null    int64
9   Month                593 non-null    object
10  Day                  593 non-null    object
dtypes: datetime64[ns](1), int64(5), object(5)

```

Image 1.a Meta Data of Sales Data

```

Data columns (total 9 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Date                 593 non-null    datetime64[ns]
1   Item Name            593 non-null    object
2   Category             593 non-null    object
3   Price                593 non-null    int64
4   Quantity             593 non-null    int64
5   Procurement Cost     593 non-null    int64
6   Week No              593 non-null    int64
7   Month                593 non-null    object
8   Day                  593 non-null    object
dtypes: datetime64[ns](1), int64(4), object(4)

```

Image 1.b Meta Data of Purchase Data

Metadata of the Columns:

1.Date:

The date on which that particular purchase/sales transaction was made

2.Item Name:

The name of the item that was sold or procured.

3.Category:

Specifies whether the product is a grocery or non-grocery item.

4.Price:

The selling price of the product per unit.

5.Quantity:

The number of units sold or procured.

6.Total Cost:

The total value of the transaction, calculated as $\text{Price} \times \text{Quantity}$.

7.Procurement Cost:

The cost incurred for procuring the goods.

8.Profit/Loss:

The difference between the selling price and the procurement cost.

9. Transaction Type:

Specifies whether the transaction was conducted in "cash" or "credit."

10.Month :

The particular month in which the transaction was made

11.Week :

The day of the week on which the purchase or sales transaction was made.

c.Data Preprocessing:

The data required for analysis was collected over a 3-month period from the business. As the store did not maintain an official ledger, all transaction details were manually recorded in a notebook. These entries included details such as date, item name, item type, quantities, and total sales amounts.

The handwritten entries were transcribed into a digital format using Google Sheets. The data had numerous redundant rows with the same date and timestamp. Those rows were deleted for better identification of unique transactions.

After the data was cleaned, it was aggregated on a weekly basis to provide a more comprehensive view of sales trends over time. It helps to understand the identification of patterns, such as peak sales periods and variations in product demand in a more effective way. By summarizing the data in this manner, the analysis became more effective, enabling meaningful insights into business performance. Part of the analysis is made in Colab whereas the rest were done with Tableau.

3.Results and findings from the analysis

A.Descriptive Analysis

Descriptive analysis is a crucial step in the data analysis process. It involves calculating measures of central tendency and dispersion to understand the data, and histograms to visualize the data, and conducting data analysis to gain insights into the characteristics of the data.

	Price	Quantity	curement (	Week No
count	593	593	593	593
mean	232.7656	2.337268	218.2125	33.18887
min	40	1	35	27
25%	50	1	40	30
50%	150	2	135	33
75%	500	3	450	36
max	600	4	600	40
std	224.4402	1.149726	209.9192	3.79029

Image 2.a Descriptive Statistics of the Purchase Data

The dataset has 593 records and includes three important variables: Price, Quantity, and Procurement Cost. The Price column has an average of 232 rupees, with a standard deviation of 224.44, indicating a significant variation in pricing. The average Procurement Cost is 218 rupees with a Standard Deviation of 209.91 rupees. The Quantity has an average of about 1.97, with a standard deviation of 1.15 indicating most of the sales are made in small quantities. The data indicates that while higher-priced products contribute to the upper quartile, reaching more than 500 rupees, lower-priced products are more frequently sold, with the median price at 150 rupees

The Procurement Cost has a mean of 270.88 and a standard deviation of 212.38, and the highest costs reaching 600 rupees. The 25th percentile for both Price and Procurement Cost is 50 and 40, respectively, whereas the 75th percentile reaches 500 and 450, suggesting a positively skewed distribution. The Quantity distribution remains more uniform, with 1 as the 25th and 50th percentiles and 3 as the 75th percentile. This summary suggests that the data exhibits a broad range of values, particularly in pricing and procurement costs, with some high-value outliers. The relationship between price, procurement cost, and quantity sold is not absolute, meaning that additional factors such as product demand, seasonal trends, and marketing strategies also play a role in sales performance.

	Price	Quantity	Total Cost	Profit/Loss	Week No
count	8	8	8	8	8
mean	257.0266	111.6744	651.2972	93.60599	125.4174
std	213.2067	207.4947	763.8685	277.693	199.6403
min	8	1	8	-400	3.79029
25%	103.75	1.904549	145	21.6875	23.9375
50%	230.5934	5.625	594.0007	63.55649	35.04722
75%	354.8943	109.1839	671.0572	159.945	124.5499
max	600	593	2400	593	593

Image 2.b Descriptive Statistics Of Sales Data

The descriptive statistics of the sales data shows that the average sale value of 257.81 rupees with a standard deviation of 213.20 rupees. The distribution business generates the most sales in July followed by August and September.

The sales trend shows fluctuations, with lower sales noted during certain periods, possibly due to seasonal variations and shifts in customer demand. External factors such as product pricing, availability, and marketing strategies also influence sales performance. While there is a general pattern in sales trends, the correlation is not absolute, indicating that other market dynamics, competition, and economic conditions also play a role in estimating sales outcomes.

B. Time - Series Analysis

Time series analysis is a powerful technique for analyzing data over time. It involves analyzing the data over time to identify trends, patterns, and seasonality. We need to understand which items are performing better to get a better picture about the shop's needs.

For time series analysis, I took the sales data since it has the records of all the items. First, I split the items based on the day of the week when the sales were made. Doing so gave a clear picture of when the item would have a demand.

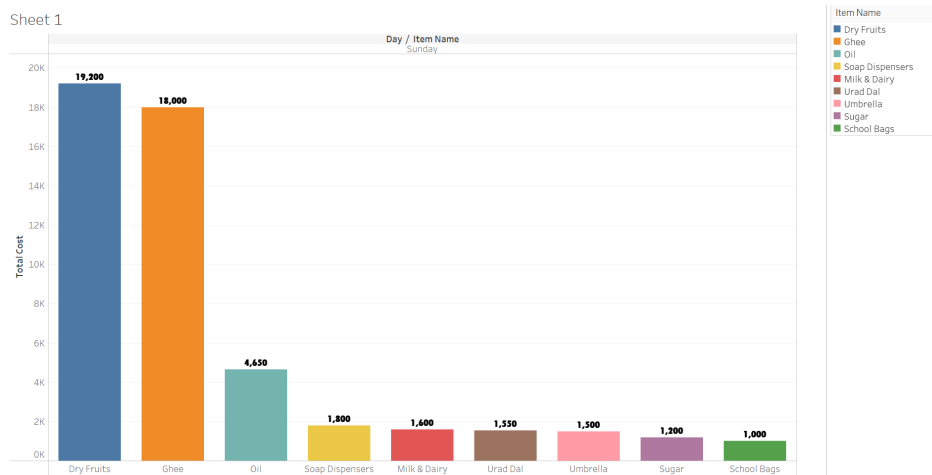


Image 2.a Items sold from the business on Sundays

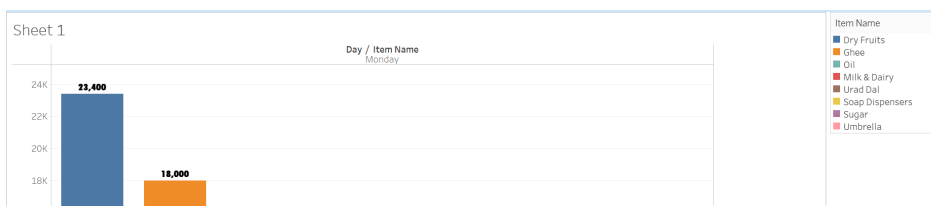


Image 2.b Items sold from the business on Mondays

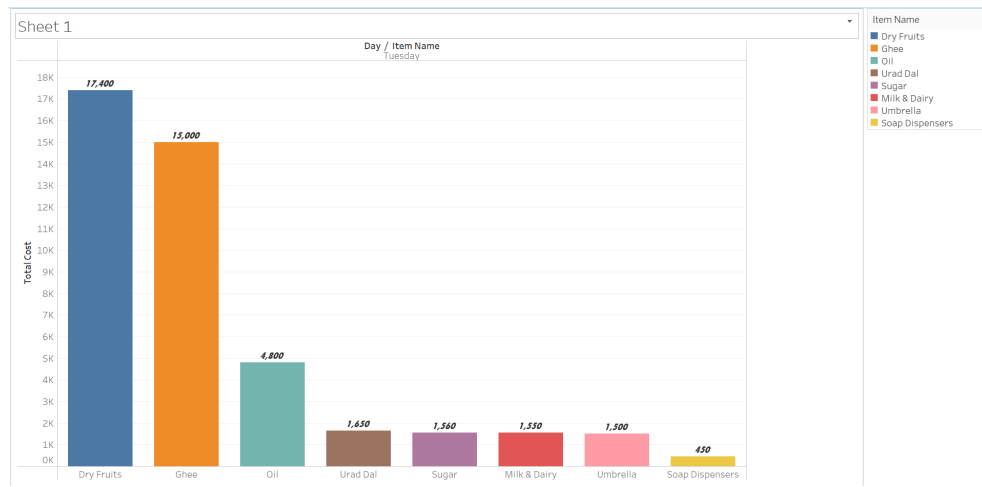


Image 2.b Items sold from the business on Tuesdays

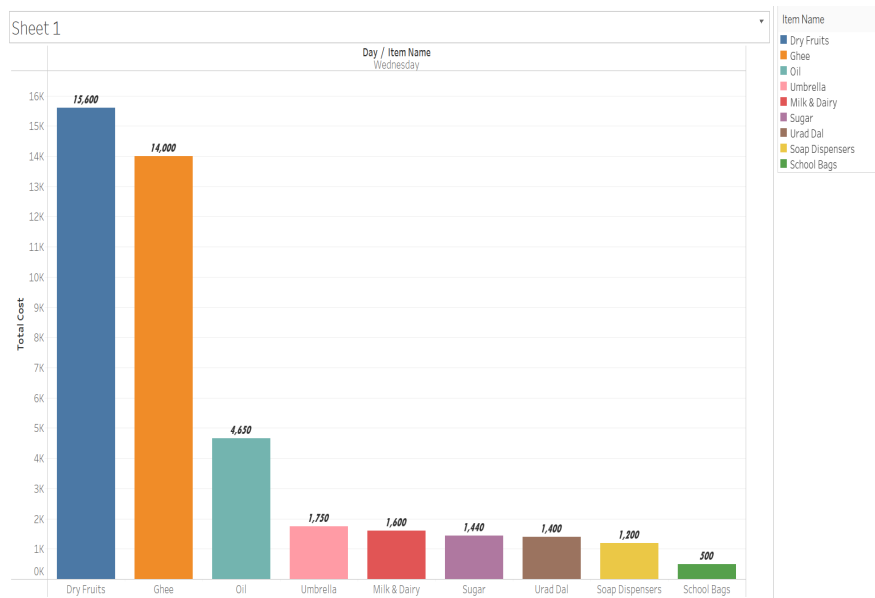


Image 2.b Items sold from the business on Wednesdays

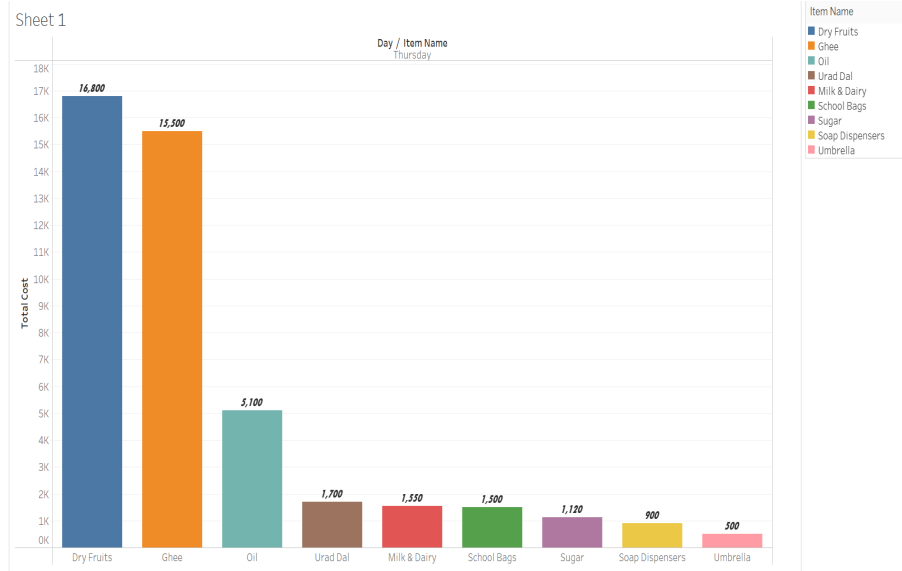


Image 2.b Items sold from the business on Thursdays

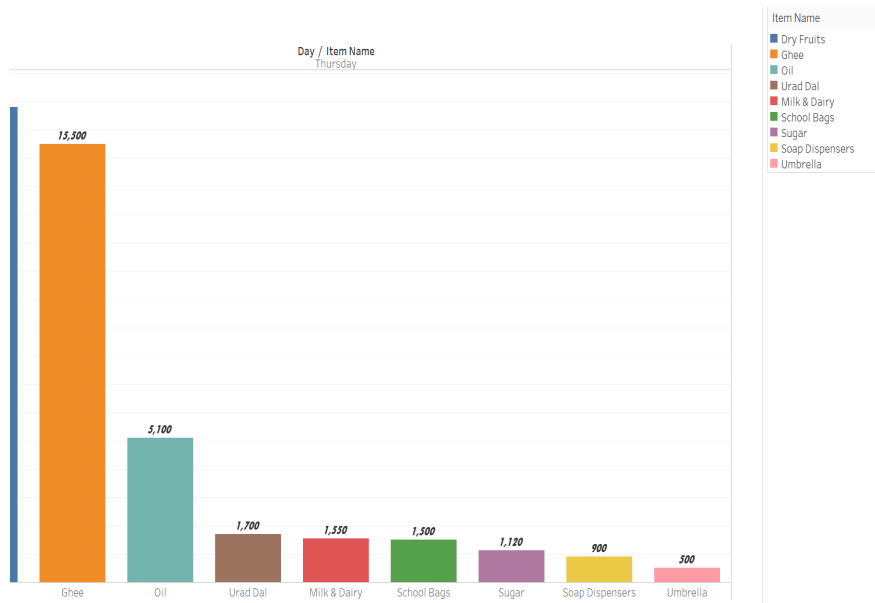


Image 2.b Items sold from the business on Fridays

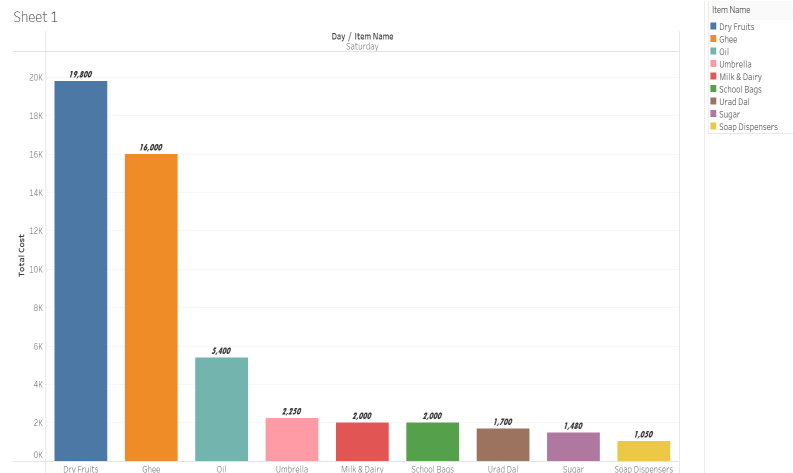


Image 2.b Items sold from the business on Saturdays

From the time series analysis , we can infer the following points

1. The most sales happen through cash transactions. Customers buy in smaller quantities, leading to high transaction volume.
2. Dry Fruits and Ghee are consistently the top-selling products, contributing significantly to total revenue. Other products like Oil, Milk & Dairy, and Sugar show steady sales, while items like School Bags, Umbrella and Soap Dispensers have lower sales.
3. Since sales occur daily, goods must be transported everyday, leading to an increase in logistics costs. Distributing goods daily requires additional manpower for handling. This increases miscellaneous expense, making cost of optimization important.

C. Purchase and Sales Data Analysis:

Analysis of sales and purchase data is an integral part of the analysis which helps to understand the correlation between the purchase made by the business and the sales done by the business.

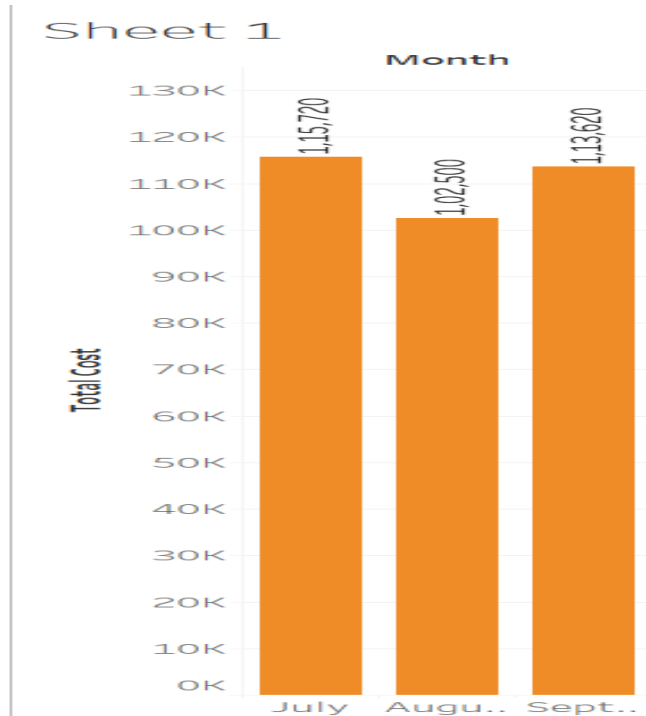


Image c.1 Graph showing the sales records for the months July, August, September

- 1.The total cost is highest in July, followed by September and August. This is likely because at the start of the quarter, the store is restocking inventory, leading to higher purchases.
- 2.The total cost then declines in August, possibly due to lower demand or fewer purchases of high-cost items. However, the cost rises again in September, indicating another round of inventory stocking or an increase in business activity.
- 3.There is a fluctuating trend in total costs rather than a steady increase or decrease. This suggests that while procurement is somewhat consistent, there are external factors influencing purchasing decisions, such as product demand, seasonal effects, or stock availability.
- 4.The purchase graph and the sales graph may show a correlation, indicating that when procurement costs rise, sales might also increase due to higher stock availability.

However, factors like unsold inventory and product demand fluctuations should be considered.

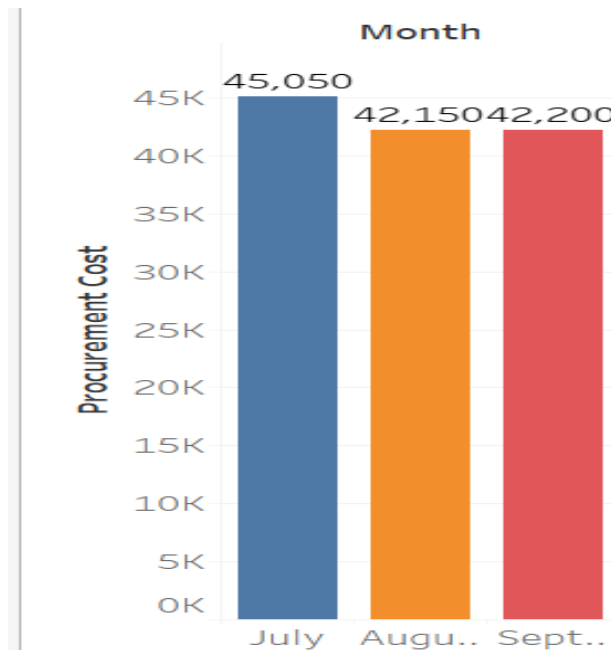


Image c.2 Image showing the purchase records for the months July, August and September

1. The relationship between the month and the total sales is not perfect. That indicates that there are other factors that affect the total sales, such as the price of the product, the availability of the product, and the marketing of the product.
2. The total procurement cost across all items sums up to ₹1,29,400, with grocery items contributing ₹1,14,450 and non-grocery items ₹14,950.
3. Grocery items generated a total profit of ₹31,190, while non-grocery items resulted in a loss of ₹9,600. The overall profit is ₹21,590, indicating that grocery items are significantly more profitable.
4. School bags (-₹1,400), soap dispensers (-₹5,000), and umbrellas (-₹3,200) led to losses, reinforcing the trend that non-grocery items are a financial drain.
5. For grocery items, higher procurement cost tends to yield higher profit, but for non-grocery items, higher procurement cost is linked to negative profits.

6. The data suggests a shift away from non-grocery items like school bags and soap dispensers and a focus on high-performing grocery items to improve profitability.

From analyzing both the purchase and the sales records of the business, we can understand that

1. The distribution business generates the most profit in July, followed by August and September. This is likely due to the start of the year, when businesses are replenishing their inventory and are more likely to make purchases from distributors. This is likely due to fresh inventory stocking.

2. The distribution business's profit then steadily declines throughout the year, with the lowest profit in September. This is likely due to the holiday season, when businesses are typically less active and make fewer purchases..

3. There is no strong positive correlation between the month and the store's profit. While July recorded the highest profit, there was a decline in August (~6.8%), followed by stabilization in September. This indicates that profitability does not necessarily improve as the months progress.

4. The profit in July was the highest, suggesting that the store effectively capitalized on seasonal demand, possibly due to high sales of essential grocery items.

5. The profit in August was approximately 6.8% lower than in July, indicating a slight decline but still maintaining stable performance.

6. The profit in September remained almost the same as in August, suggesting that the store was able to sustain its sales but did not experience significant growth.

7. The decline from July to August may be due to reduced demand or higher procurement costs not translating into equivalent sales growth. The store should investigate whether stocking patterns or pricing strategies need adjustments.

8. If this pattern continues, the store may need to implement promotional strategies or optimize its inventory to avoid further decline in the following months.

D. Forecasting And Predictive Analysis:

Forecasting and predictive modeling are techniques for predicting future values of the data. They involve using statistical techniques such as regression analysis, time series forecasting, or machine learning algorithms to analyze historical data and generate predictions for future values of the data.

One of the main problems that is faced by the business is that they find it difficult to understand the demand trend of the market. Someday, the shop buys goods in large quantities whereas on some other days, they buy in small quantities.

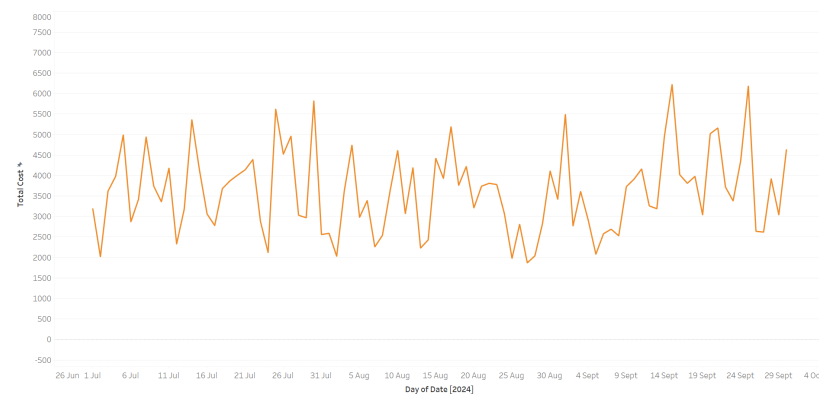


Image 4.a Graph showing the trend between the total value of goods sold vs time

From the above graph, it can be seen that the sales are at peak during particular times of the year. On further investigation, it was found that these peaks align with festive seasons. The sales in July are greater than August and September month sales.

But this behavior doesn't explain the low in the sales observed in August and September. Closely examining the timeline and the product being sold, we can see the general idea of all the households. Households may shift spending priorities during these months, focusing on school fees and other expenses.

B. Competitive Analysis:

Competitive analysis is a method used for evaluating the performance of a distribution business in comparison to its competitors. The process includes gathering data on competitors' sales and purchasing patterns, examining market share and market

trends, and identifying strengths, weaknesses, opportunities, and threats (SWOT analysis) in relation to the competition.

By conducting competitive analysis, one can gain insights into the competitive landscape and identify areas where business can gain a competitive advantage. This analysis can help you understand how your business is positioned in the market, identify potential gaps in the market that you can exploit, and develop strategies to differentiate your business from competitors.

Key components of a competitive analysis include evaluating competitors' pricing strategies, customer service, and marketing efforts. By comparing these factors to your own business, you can identify areas for improvement and develop strategies to surpass the competition.

4. Interpretation of Results and recommendations to the business

1. Reasons for having low profit even though the returns are high

The profit returns for the business is calculated as the difference between sales and purchase costs. However, in this case, profitability varies across two different categories. Grocery items generate high profits, while non-grocery items lead to loss.

The sales and purchase records for grocery items have a high correlation, indicating that the sales in a month correspond closely with the purchases made in the same month. This simplifies the process of calculating profit using the traditional formula: profit

$$\text{Profit} = \text{Sales} - \text{Purchase}$$

In addition to the purchase investment, the business also makes investments in transportation. These miscellaneous expenses are added to the calculations when calculating the profit. As a result, the equation for calculating profit changes to

$$\text{Profit} = \text{Sales} - \text{Purchase} - \text{Miscellaneous.}$$

Since the miscellaneous expenses add to the cost of goods sold, the profit is always lower than expected for the business.

2. Understanding The Demand

The demand for the products sold by the business is significantly influenced by seasonal variations. The analysis revealed that months such as July, August and September tend to show the highest demand for the products in all the years. This is largely due to the festivals that fall in these months, such as Krishna Jayanthi and Vinayagar Chathurthi. During these months, customers tend to purchase more grocery items and also puja essentials.

The analysis revealed that July tends to show the highest demand for school bags, as students prepare for the new academic year. However, school bags continue to be sold throughout July, August, and September in the store even though the actual demand is only in July. Similarly, umbrellas have the highest demand in November and December due to the monsoon season, but they are stocked and sold in July, August, and September at the store despite lower demand.

Understanding the seasonality of demand for the products is important for the business to enhance its operational efficiency and increase profitability. By anticipating periods of high demand, the business can ensure that it has sufficient inventory to meet customer needs. By anticipating periods of low demand, the business can modify its operations to reduce costs and improve efficiency. By aligning its operations with the seasonality of demand, the business can maximize its sales and profitability and maintain a competitive edge in the market.

c. Trends of Miscellaneous expenses:

The miscellaneous expenses at Kanish Stores primarily include transportation cost and rent. These expenses are essential for the business's operations and contribute to its overall cost structure. Unlike other businesses, there are no labor expenses, as the store does not employ additional workers for operations. Rent remains a fixed and important expense that cannot be reduced, as it is required for maintaining the store's physical space.

The transport expense is an important part of miscellaneous expenses. Instead of a dedicated delivery system, the shop owner personally visits different wholesale stores daily to purchase goods. This ensures that fresh stock is regularly available but also leads to high transportation costs, as the owner frequently travels to multiple locations.

Since purchases are made on a daily basis, transportation costs can become inefficient due to repetitive travel patterns. The owner often revisits the same wholesale stores multiple times a week, leading to increased fuel expenses and unnecessary travel time.

To optimize transport expenses, the business could consider planning purchases in bulk to reduce the frequency of visits. By identifying the most frequently bought products and stocking them for a longer period, the owner can minimize daily travel, leading to significant cost savings. Additionally, adopting route optimization strategies by grouping purchases from nearby wholesalers on the same trip can reduce fuel consumption and save time.

Another potential approach to managing transport expenses is to negotiate better pricing or delivery options with wholesalers. Some suppliers may offer discounts for bulk purchases or even provide delivery services, which could further reduce transportation costs. By implementing these strategies, Kanish Stores can reduce unnecessary transportation costs, improve efficiency, and enhance overall profitability.

C. Other Recommendations:

The business can encourage customers to purchase in large quantities by providing discounts or promotional deals on specific days. For example, if customers are informed about discounted prices on certain products on a fixed day each week, it could enhance sales while optimizing inventory management. This approach can also help in reducing transportation costs, as larger purchases mean fewer trips required to restock.

Additionally, reducing the profit margin on certain essential goods can attract more customers. Although this may lead to lower initial profits, an increase in customer footfall and higher sales volume will eventually enhance overall profitability. By strategically lowering prices on high-demand items, the store can gain customer loyalty, leading to repeat purchases and consistent revenue.

The decrease in profit margin on selected products can also encourage bulk buying, leading to faster inventory turnover. Customers may prefer Kanish Stores over competitors due to favourable pricing, thereby increasing overall sales and compensating for the lower margins through increased volume.

